

Influence of Nanoporous Carbide-Derived Carbon Electrode Morphology on Charge Accumulation in Supercapacitors

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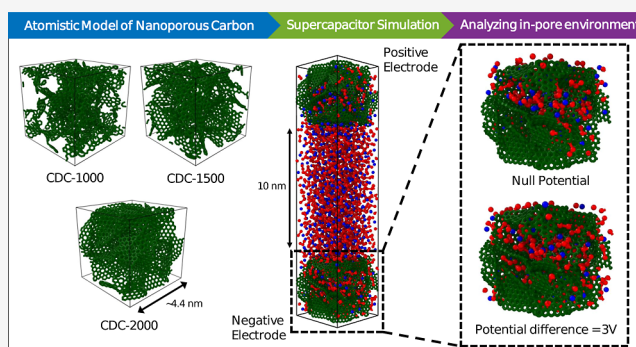
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ABSTRACT: Electrochemical devices that utilize ionic liquids and porous carbon electrode materials have garnered significant interest due to their promising characteristics. A thorough understanding of the charging mechanisms at the molecular level is essential to enhancing energy and power densities simultaneously. In this work, we simulated supercapacitors using realistic atomic models of electrodes and a room-temperature ionic liquid (RTIL) to investigate how the local environment within the electrodes affects the charge accumulated on the electrode atoms. Three atomistic models with realistic structural features of experimentally synthesized carbide-derived carbon (CDC) at different temperatures were generated following the annealing simulation methodology with an analytic bond-order potential (ABOP) force field. It was observed that the pore size distribution (PSD) of the CDC electrodes shifted to the right as the synthesis temperature increased. The results obtained from simulating supercapacitors with these electrode models indicated that counter-ions are strongly confined and isolated from co-ions in electrodes with smaller average pore sizes in supercapacitors than those with larger average pore sizes. This finding was supported by changes in the profiles of the coordination number and degree of confinement (DoC). However, electrodes with small pores limited the number of counter-ions that could be accommodated. In contrast, electrodes with larger average pore sizes permitted more counter-ions and facilitated faster diffusion within the electrode. As a result of these opposing trends—confinement and the number of adsorbed counter-ions—there was a peak in the accumulated charge for the electrode with an intermediate average pore size.



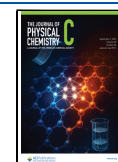
1. INTRODUCTION

In recent years, supercapacitors have emerged as novel energy storage devices, offering higher power density than batteries.¹ They have been utilized in various power system applications, such as maximum power point tracking in photovoltaic systems,² energy intermittency mitigation,³ and power generation smoothing⁴ in renewable energy sources. Supercapacitors have significantly enhanced transportation technologies due to their rapid charging and discharging time. They are particularly valuable in situations where high current is needed for a short duration, such as during regenerative braking, acceleration, vehicle starting, and stopping.⁵ Additionally, supercapacitors find application in hand-held electric appliances and power electronics and as a backup for memory chips.⁶ The diverse range of applications can be attributed to the advantageous features of supercapacitors, including a high power density of 196 kW/kg⁷ and an extended cyclic stability of 30,000 to 1,000,000 h,⁸ which is orders of magnitude greater than conventional battery technologies. However, despite these benefits, the widespread adoption of supercapacitors is limited by their low energy density relative to battery technology.⁹

The advantages of supercapacitors have spurred extensive research to create devices with a higher energy density. Various

strategies have been investigated to enhance the energy density, including using electrolytes with high electrochemical stability potential windows,¹⁰ developing novel electrode materials,^{11,12} and designing asymmetric structures.¹³ Carbon-based electrode materials are promising for electric double-layer capacitors (EDLC) due to their cost efficiency, high surface area, and tunable pore size distribution (PSD). In EDLCs, charges accumulate as liquid electrolyte ions are adsorbed at the electrodes and electrolyte interfaces, creating alternating counter- and co-ion layers.¹⁴ The high capacitance value was initially attributed to the high surface area of porous carbon structures, with the pore sizes playing a significant role.¹⁵ The capacitance normalized by the surface area increases considerably if the pore size matches the electrolyte ion size, where the solvation shell of electrolyte ions gets distorted and ions can

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closely approach the pore walls inside subnanometer pores.¹⁶ The cumulative effect of high surface area and subnanometer pore size made carbide-derived carbon (CDC) porous materials a major focus in developing supercapacitor (SC) devices with high energy density.

Significant advances have been made in the experimental study of supercapacitors.^{17,18} A nuclear magnetic resonance analysis was conducted to quantify the changes in the number of counter- and co-ions inside micropores of activated carbon (AC) electrodes and ionic liquid electrolytes.¹⁹ The findings indicated that anions played a dominant role in the charge storage mechanism despite their volume, similar to that of cations in the studied electrolyte. Factors such as binding energy, steric effect, and charge distribution of the ions may explain these differences, and varying charging mechanisms may be observed for different electrolytes with the same carbon electrode. The experimental data were interpreted using computer simulations to explain the ion positions in AC and CDC electrodes under applied potential.²⁰ A three-dimensional real-space representation of the porous electrode and the global concentration of electrolytes was obtained from small-angle X-ray scattering. In computer simulations, ions were arranged within the electrode using a Monte Carlo algorithm to emulate the equilibrium state of the supercapacitor under applied voltage.²⁰ Theoretical and simulation analyses are indispensable for interpreting the experimental data and molecular-level interactions.

EDLC devices have been extensively studied by molecular dynamics (MD) simulations, which have explored various types of electrodes, including planar graphite,²¹ slit pore,²² and zeolite-based electrodes.²³ However, these numerical simulations face significant limitations in attempting to model realistic scenarios. First, the study of EDLC devices incorporating porous carbon electrodes has been limited, primarily due to the challenges in creating an atomistic model of a realistic amorphous CDC porous material. Most investigations of porous carbon electrodes have utilized a slit pore model, which consists of two planar graphene sheets separated by a fixed distance that corresponds to the peak pore size of an experimental sample.²⁴ This simplistic model does not accurately reflect the amorphous nature of the actual CDC materials. Although researchers have generated more complex geometric structures to simulate realistic CDC materials,^{25,26} these models still struggle to capture the intricate pore architecture present in real materials. Second, modeling the experimental conditions where a constant voltage is applied to the electrodes is both algorithmically complex and computationally expensive. Consequently, some MD simulations of supercapacitors have adopted the constant charge method in which the charge of the electrode atoms is set manually at the beginning of the simulation. The voltage is then calculated after equilibrating the system to calculate the capacitance.²⁷ However, this approach diverges from the actual experimental process, leading to results that significantly differ from the real-world behavior.²¹

Several studies have been reported in the literature that focus on simulating supercapacitors by using an atomistic model of the CDC material and the constant potential method (CPM). The CDC model developed in the quench MD method by Palmer et al.²⁸ was integrated with an ionic liquid electrolyte to simulate supercapacitors in several studies.^{29–31} However, when comparing the radial distribution function (RDF) of the simulated atomistic model with the experimental X-ray diffraction (XRD) pattern, a significant mismatch was

discovered.³² Recently, a CDC structure generated using the ReaxFF force field³³ was employed to simulate EDLC devices,³⁴ although the CDC structure was not realistic.³⁵

Despite the discrepancies from experimental structures, the CDC atomistic models developed by Palmer et al.²⁸ were the most commonly used materials for porous electrodes in supercapacitor studies. Recently, new atomistic models of CDC structures have been generated with an analytic bond-order potential (ABOP)³⁶ in the annealing method, which closely mimics the experimental process of producing CDC materials.³⁷ The concept of temperature acceleration was applied to simulate the reaction dynamics within the time scale accessible by MD simulation.³⁸ The temperature variations of various structural parameters in the simulated models closely mirrored those observed in the experimental CDC samples. Additionally, the RDF profile showed a strong alignment with experimental XRD patterns across a wide range of synthesis temperatures, representing a significant improvement over the previous models by Palmer et al. Therefore, it is essential to simulate supercapacitors by using realistic atomistic models of electrodes to achieve a more accurate representation of real-world conditions.

This work simulated EDLC devices using realistic nanoporous CDC electrode models combined with room temperature ionic liquid (RTIL) 1-butyl-3-methylimidazolium hexafluorophosphate (BMIM⁺PF₆[−]) through a constant potential method. We calculated the accumulated charge and specific capacitance for these electrodes and observed significant variations in these parameters, depending on the synthesis temperature of the electrodes. Several parameters depicting the environment surrounding the adsorbed ions in porous electrodes were calculated, and their variation was observed for different electrode types. The electrodes included in this investigation corresponded to experimental samples synthesized at 400, 600, and 800 °C. The PSD shifted progressively to the right with an increase in temperature, highlighting two competing trends affecting the accumulated charge for the electrodes. The electrode with a smaller average pore size was capable of confining ions more effectively while also having a higher inaccessible volume. The more confined ions could induce a greater charge. Conversely, a less inaccessible volume in electrodes with a left-shifted PSD allowed fewer ions to be adsorbed. To gain insight into the impact of the porous material on the arrangement of adsorbed ions, we compared parameters that characterize the local environment of electrolyte ions across the three electrode types.

2. SIMULATION METHODOLOGY

The electrodes of the supercapacitor were generated in the annealing methodology with the ABOP force field.³⁷ The atomistic models of porous carbon generated in this method are the most accurate representation of CDC nanoporous structures, to the authors' knowledge. Three different types of electrodes were generated at 1000, 1500, and 2000 K, corresponding to experimental synthesis temperatures of 400, 600, and 800 °C, respectively. For convenience, these electrode types will be referred to by their simulation synthesis temperatures as CDC-1000, CDC-1500, and CDC-2000 in subsequent discussions. Details regarding the PSD, surface area, volume, average pore width, and diameter of the largest free sphere of the electrodes can be found in the [Supporting Information](#). Notably, the PSD shifted to the right as the synthesis temperature for the electrode increased, as shown in

Figure S1. The rightward shift is evident from Table S1 as the average pore diameters for CDC-1000, CDC-1500, and CDC-2000 were 7.16, 7.79, and 9.86 Å, respectively.

The cubic CDC structures are ~4.4 nm long in each dimension. The supercapacitor consisted of two electrodes separated by a distance of 10 nm in the *z* direction, as shown in Figure 1. The right electrode in Figure 1 was inverted along the *z*

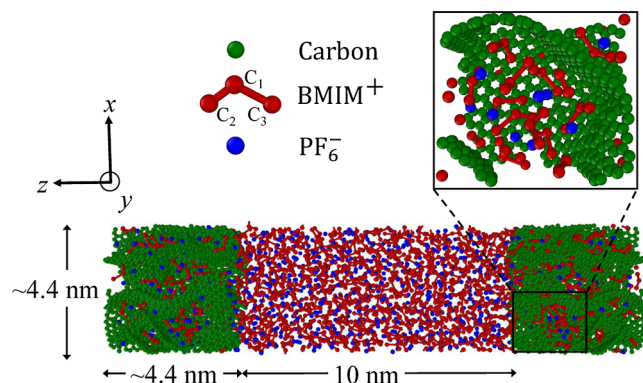


Figure 1. Schematic diagram of the supercapacitor with CDC-2000 electrodes under applied voltage. The applied potential is positive on the left electrode. The inset exhibits a cross-sectional view near the interface between the electrode and electrolyte of the negative electrode, showing electrolyte ions adsorbed within pores. The green, red, and blue colors represent carbon atoms, BMIM⁺ ions, and PF₆[−] ions, respectively. The coarse-grained model of the cation (BMIM⁺) consisted of three sites denoted by C₁, C₂, and C₃.

axis, ensuring that the same surface of both electrodes faced the bulk electrolyte in the EDLC device. The void space between the electrodes was filled with a random arrangement of an equal number of BMIM⁺ and PF₆[−] electrolyte ions. The number of electrolyte ions was selected from several trial runs, as given in Table S2 of the Supporting Information, such that the bulk electrolyte density at null potential was close to the experimental density of the ionic liquid at 400 K. The interaction between electrode and electrolyte ions was modeled by Coulombic pairwise interaction along with the 12–6 Lennard-Jones (LJ) potential³⁹ as given by

$$E_{\text{LJ}} = 4\epsilon \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right] \quad r < r_c \quad (1)$$

In eq 1, ϵ , σ , and r_c are the energy, size parameters, and cutoff radius, respectively. The cross parameters are calculated using the Lorentz-Bertholet mixing rule. The parameter values are given in Table 1.

Table 1. Force Field Parameters of the Electrolyte and Carbon Atom^a

	mass (g/mol)	σ (Å)	ϵ (kcal/mol)	q (electron charge)
C ₁	67.07	4.38	0.609524	0.4374
C ₂	15.04	3.41	0.085714	0.1578
C ₃	57.12	5.04	0.435714	0.1848
A	144.96	5.06	1.12143	−0.78
carbon	12.0107	3.37	0.05497	N/A

^aThe parameters are collected from the COMPASS force field⁴⁰ and the work of Roy and Maroncelli.⁴¹ The coarse-grained model of the cation consists of three sites represented by C₁, C₂, and C₃. The anion consists of only one site (A).

The electrode–electrolyte systems were simulated using the LAMMPS software package.⁴² Throughout all stages of the MD simulation, the positions of carbon atoms in the electrodes were kept fixed, allowing only the electrolyte ions to move. Short-range electrostatic and LJ interactions were calculated within a cutoff radius of $r_c = 1.5$ nm. For long-range electrostatic interaction beyond this cutoff, the particle–particle/particle–mesh (PPPM) method was employed.⁴³ The supercapacitor was modeled with periodic boundary conditions applied in the *x* and *y* directions, while fixed boundary conditions were used in the *z* direction. The nonperiodic boundary condition in the *z* direction, combined with the PPPM method, required the removal of interslab dipole interactions to ensure an accurate simulation. The dipole interaction was effectively turned off by extending the *z*-dimension by a factor of 3.⁴⁴ The angle and bond lengths of the BMIM⁺ ion were kept constant by the SHAKE algorithm.^{45,46} The velocity-Verlet algorithm was employed to integrate the equations of motion with a time step of 2 fs.

The initial structure's energy was minimized using the conjugate gradient scheme, with a force and energy tolerance of 10^{-8} and 10^{-8} eV/Å. The system was then simulated at 400 K for 10 ns in the canonical (NVT) ensemble, with the Nose–Hoover thermostat's temperature relaxation constant set to 10 ps. In the following 5 ns, both electrodes were held at zero potential to allow the system to equilibrate. Voltage was applied to the electrode atoms using the CPM⁴⁷ by incorporating the LAMMPS ELECTRODE package.⁴⁸ The only adjustable parameter of the CPM, i.e., the width of the Gaussian charge distribution, was set to 0.5052 Å.⁴⁹ During the final stage of the equilibration run, a potential difference of 3 V was applied between the electrodes, and this stage lasted until the charge accumulation on the electrodes approached saturation. Approximately, the charging stage would last for tens of nanoseconds. Throughout the last 10 ns of the simulation, the coordinates and charges of all of the atoms were recorded every 5 ps. The steady-state coordination number distribution, degree of confinement (DoC), minimum distance distribution, and atomic charge distribution were calculated by averaging the data collected from this final 10 ns period.

The local environment was characterized by the coordination number, DoC, and distance of the closest approach (d_c) calculated for the counter-ions. These parameters were averaged over the total number of counter-ions adsorbed and averaged over 2000 frames extracted from the last 10 ns of the simulation. On the other hand, the parameters corresponding to the null potential equilibration period were calculated by averaging over 500 frames taken from the last 2.5 ns of the equilibration period at zero potential difference. The coordination number refers to the number of co-ions surrounding a counter-ion. The coordination number profile for counter-ions inside the electrodes is calculated to understand the influence of the electrodes' nanoporous morphology on charge accumulation. The distance of the closest approach is the smallest distance between the center of mass of electrolyte ions and the electrode atoms. The DoC represents the percentage of solid angle occupied by electrode atoms around the adsorbed ions, normalized by its maximum value. This concept was first described by Merlet et al.³⁰ However, the formula used by Merlet et al. to compute the solid angle corresponding to individual carbon atoms was not applicable to spherical geometries, i.e., atoms. Furthermore, the ions were found to adsorb in pores with diverse structural characteristics, which

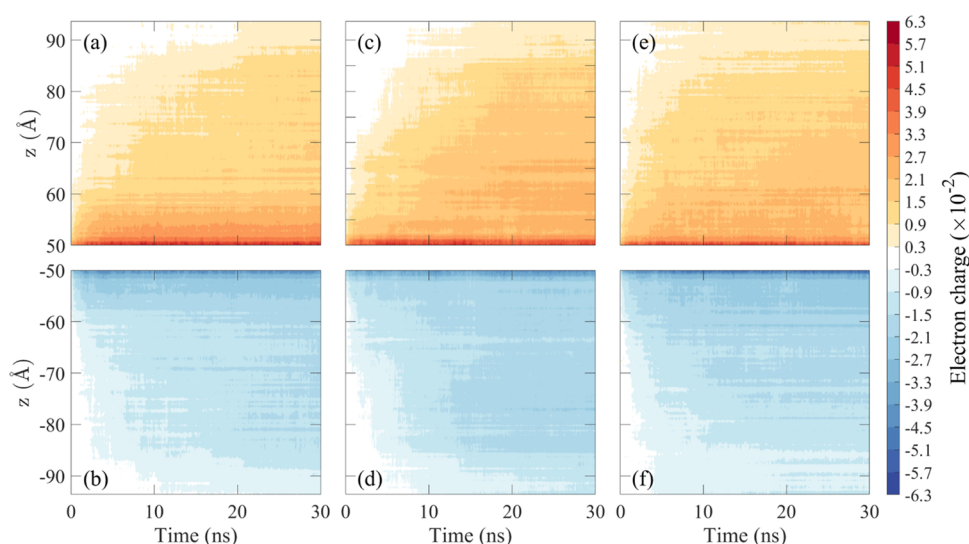


Figure 2. Variation of average charge per carbon atom with time and distance, for the first 30 ns of the simulation, along the z -axis in (a,c,e) positive and (b,d,f) negative electrodes for (a,b) CDC-1000, (c,d) CDC-1500, and (e,f) CDC-2000 electrodes under an applied voltage of 3 V between the electrodes.

occasionally led to the overlapping of solid angles traced to electrolyte atoms for multiple electrode atoms located within the sphere of influence defined by the cutoff radius.

To address these challenges, this work utilized the Monte Carlo method to accurately calculate the solid angle occupied by electrode atoms, as detailed in the [Supporting Information](#). The number of random points used for calculating the solid angle for each ion was set at 10^5 to ensure that the percentage deviation between actual and calculated values was less than 1% (as shown in test cases within the [Supporting Information](#)) while maintaining a reasonable computation time. The DoC was calculated from the solid angle following the formula³⁰

$$\text{DoC} = \frac{\Omega_i}{4\pi \times 0.6046} \times 100 \quad (2)$$

In eq 2, Ω_i is the solid angle corresponding to the i th ion.

Calculating the coordination number and DoC required definition of a sphere of influence surrounding the ions. The radius of the sphere corresponding to counting the coordination number was determined from the first minima of RDF calculated between the counter-ions and co-ions. The first minima of RDF of counter-ions and electrode atoms was the sphere's radius for calculating DoC. The cutoff radii for calculating the coordination number and DoC were 8.0 and 8.5 Å, respectively.

3. RESULTS

At an applied voltage, the anion (cation) and cation (anion) were adsorbed and desorbed in the positive (negative) electrode. Ions that have opposite polarity relative to the electrode potential are referred to as counter-ions. In this context, PF_6^- and BMIM^+ are considered counter-ions at the positive and negative electrodes, respectively. Conversely, BMIM^+ and PF_6^- ions act as co-ions in the positive and negative electrodes. The anion BMIM^+ has three active sites, and the positions of these ions are determined by their centers of mass.

3.1. Transient Behaviors under Applied Voltage. The transient behavior of the average charge on electrode atoms in the z direction, relative to the electrode–electrolyte interface for different electrodes, is illustrated in the contour plot shown in [Figure 2](#). The average charge per carbon atom was calculated by

considering several slices of the electrode along the z direction and dividing the amount of charge by the number of atoms within each slice. The profiles revealed some common characteristics in the charging patterns of the electrodes used in supercapacitors. First, the atoms located near the electrolyte–electrode interface are charged quickly, reaching a high average value of $\sim 5.5 \times 10^{-2}$ electron charge (e). They maintained this charge for the duration of the simulation. As time progressed, carbon atoms situated deeper within the electrode also became charged, eventually achieving a steady-state value. This pattern was anticipated, as the electrolyte ions required time to navigate the complex channels to reach the deeper pores and induce charge on the electrode atoms. The steady-state accumulated charge value diminished rapidly from the interface into the deeper regions of the electrodes.

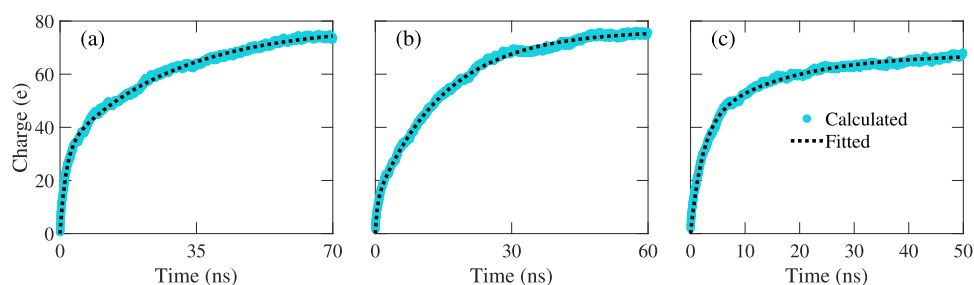
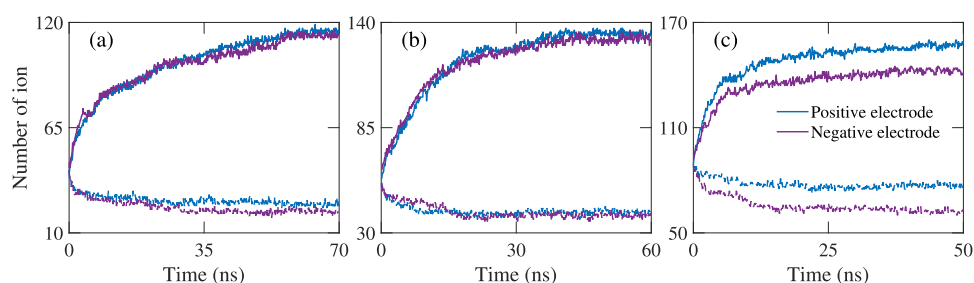
The most noticeable difference among the electrode types was the speed at which the charge was induced from the bulk electrolyte–electrode interface. The atoms located at $z = 94$ nm required ~ 20 ns to acquire an average charge of $>0.03 e$ in the CDC-1000 electrode, as shown in [Figure 2a](#). In contrast, it took only ~ 4 ns in the CDC-2000 electrode, as illustrated in [Figure 2e](#). This observation suggests that the further right an electrode's PSD shifts (depicted in [Figure S1](#)), the faster ions can diffuse through the structure and induce charges on deeper atoms. Another noteworthy observation from [Figure 2](#) is that the profile of the average charge per atom follows an exponential-like trend over time for all types of electrodes, indicating that the ions diffuse rapidly into the electrodes at the beginning of the voltage application period. However, the ions required significantly more time to penetrate the deepest pores, which is evident in [Figure 2d](#). The atoms located in the range $-50 > z > -85$ nm accumulated an average charge of $<-0.09 e$ at ~ 8 ns. Beyond this point, the deeper atoms picked up charge much more slowly.

The charging dynamics of supercapacitors with nanoporous electrodes differs significantly from those of conventional capacitors. This discrepancy is highlighted by the poor fit of the transient charge to a single exponential equation.²³ The underlying cause of this phenomenon is the dynamic heterogeneity of the ionic liquid, which leads to variations in the capacitance of supercapacitors during the charging process.

Table 2. Calculated Parameters for CDC-1000, CDC-1500, and CDC-2000 Electrodes^a

electrode type	$Q_{\text{fit}}(e)$	c	τ_1	τ_2	steady-state charge, $Q_{\text{ss}}(e)$	specific capacitance, $C_g(\text{F/g})$
CDC-1000	78.3	0.603	28.2	1.48	74.0	98.6
CDC-1500	76.5	0.844	15.13	0.45	75.25	100.3
CDC-2000	67.6	0.596	2.19	15.7	67.4	90.4

^aThe parameters Q_{fit} , c , τ_1 , and τ_2 were extracted from fitting eq 3 to total accumulated charge variation with time. The specific capacitance C_g was calculated using eq 4.

**Figure 3.** Evolution of total accumulated charges on the positive electrode with 3 V potential difference between the electrodes for (a) CDC-1000, (b) CDC-1500, and (c) CDC-2000, respectively. The dotted line shows the fitted curve drawn with eq 3.**Figure 4.** Variation in the counter- and co-ion numbers within positive and negative electrodes for (a) CDC-1000, (b) CDC-1500, and (c) CDC-2000 electrodes. The continuous and broken lines represent the counter- and co-ions, respectively.

To model the variation of electrode charge over time in the supercapacitor, we employed two different equations: the stretched exponential function and the biexponential function.⁵⁰

Our study used the biexponential expression because it yielded a smaller root mean squared error than the stretched equation. The equation is as follows

$$Q(t) = Q_{\text{fit}}[1 - ce^{-t/\tau_1} - (1 - c)e^{-t/\tau_2}] \quad (3)$$

where Q_{fit} , c , τ_1 , and τ_2 are the fitting parameters.

The dominant exponential term of eq 3 can be identified from the value of c . In each case, $c > 0.5$, indicating that the exponential factor with τ_1 is more significant. The fitting parameters for eq 3 are given in Table 2. By comparing the τ_1 values of the electrodes, we gain a quantitative understanding of their transient behavior. The value of τ_1 decreased from 28.2 to 2.19 for CDC-1000 to CDC-2000, which implies that the CDC-2000 electrode accumulated charge much faster and reached a steady state sooner than the other two electrodes, as illustrated in Figure 3 and evidenced by the smaller τ_1 value. This observation aligns with the findings in Figure 2, which indicate that even the deepest pores became charged rapidly for the CDC-2000 electrode. The steady-state charge (Q_{ss}) was calculated by averaging the charge of the electrode over the last 10 ns of the simulation.

The specific capacitance (C_g) for each electrode was calculated by³¹

$$C_g = \frac{2Q_{\text{ss}}}{m\Delta\phi} \quad (4)$$

In eq 4, m and $\Delta\phi$ are the mass of the electrode and the potential difference between the positive and negative electrodes, respectively. The C_g value was a maximum for the CDC-1500 electrode, as shown in Table 2. The C_g initially increased and then went down with respect to the synthesis temperature, as shown in Table 2, where the maximum $C_g = 100.3 \text{ F/g}$ for CDC-1500. The average pore size for the CDC-1500 structure was $9.1 \text{ \AA}$.³⁷ The obtained C_g is comparable to the previously reported value of 125 F/g for supercapacitors, with the porous CDC electrode having an average pore size of $9.45 \text{ \AA}$.^{28,29} However, the electrode model in ref 29 used the Reactive Summation State carbon force field⁵¹ and differed significantly from the physical CDC structure.

The value and trend of C_g with average pore size observed in this work was similar to the experimental work studying supercapacitor devices with TiC-derived nanoporous electrodes synthesized at different temperatures ranging from 500 to 1000 °C¹⁵ and tetraethylammonium $[\text{TEA}^+]$ tetrafluoroborate $[\text{BF}_4^-]$ in acetonitrile. The electrodes CDC-1000, CDC-1500, and CDC-2000 corresponded to experimental samples generated at the synthesized temperatures of 400, 600, and 800 °C.³⁷ The C_g values for electrodes synthesized at 500, 600, and 800 °C were 140, 142, and 130 F/g.¹⁵ The maximum value appears at 600 °C, similar to CDC-1500. The percentage change from 600 to 800 °C was the same as that for CDC-1000 to CDC-1500. The

Table 3. Numbers of Adsorbed Counter- and Co-Ions inside Positive and Negative Electrodes, Averaged over the Last 10 ns, for Supercapacitor (SC) Devices with CDC-1000, CDC-1500, and CDC-2000 Electrodes at Null and 3 V Potentials^a

electrode type	electrode potential	$N_{\text{counter}}(3)$	$N_{\text{co}}(3)$	$N_{\text{counter}}(0)$	$N_{\text{co}}(0)$	$N_{\text{counter}}(3) - N_{\text{co}}(3)$	$X(3, 0)$
CDC-1000	positive	112.7	25.7	40.7	41.7	87.0	0.64
	negative	110.3	21.0	42.2	42.0	89.3	0.53
CDC-1500	positive	134.0	40.9	56.4	53.7	93.1	0.72
	negative	131.5	39.2	57.9	57.0	92.3	0.61
CDC-2000	positive	157.8	76.75	89.8	89	81.1	0.69
	negative	142.5	61.6	88.9	85.9	80.9	0.37

^aThe charging mechanism parameter X was calculated using eq 5.

diameters of electrolyte ions $[\text{TEA}^+]$ and $[\text{BF}_4^-]$ were 6.8 and 3.3 Å, respectively. On the other hand, the maximum dimensions of $[\text{BMIM}^+]$ and $[\text{PF}_6^-]$ were 9.6 and 5.06 Å, respectively. The larger sizes of the pairs of cation and anion compared to the ions employed in the experimental study are responsible for the lower value of simulated C_g compared to that of the experimental data.

The variation in the number of ions within the positive and negative electrodes is illustrated in Figure 4. Initially, the CDC-1000 and CDC-1500 electrodes were partially filled with ions. Upon applying voltage, a significant number of counter-ions entered pore volumes that had previously been unoccupied in both the negative and positive electrodes. Consequently, the ion number variation over time was similar for both electrodes in Figure 4a,b. In contrast, the ion number variation for the CDC-2000 electrode differed significantly from those of the CDC-1000 and CDC-1500 electrodes. First, the counter-ions number increased rapidly, mirroring the swift rise in average charge per atom depicted in Figure 2e,f. This rapid change in ion number was due to the larger pore size of the CDC-2000 electrode, which facilitated the quick diffusion of ions within the electrodes. Second, the variations in counter- and co-ions differed between the positive and negative electrodes for CDC-2000, owing to its markedly right-shifted PSD, as shown in Figure S1. The largest free sphere and average pore size of the CDC-2000 electrode were 7.46 and 9.86 Å, respectively, which are significantly larger than those of CDC-1000 and CDC-1500. As a result, a substantial portion of the accessible pore volume in the CDC-2000 electrodes was occupied by ions at null potential, in contrast to the conditions observed in CDC-1000 and CDC-1500.

After the voltage was applied, co-ions were expelled from the electrodes, and counterions were admitted. The difference between the profiles corresponding to positive and negative electrodes shown in Figure 4c is due to the van Der Waals volume of BMIM^+ being almost twice that of the PF_6^- ion. The counter-ion numbers in the negative and positive electrodes were ~ 160 and 140 , respectively, and the co-ion numbers were ~ 80 and ~ 60 at the end of the simulation. Once the BMIM^+ ions moved out, a greater number of PF_6^- ions could be adsorbed in the empty place because of the volume difference of the ions. In the negative electrode, more co-ion PF_6^- needed to be pushed out to accommodate counter-ion BMIM^+ , thus resulting in the difference in trend in the counter-ion number for positive and negative electrodes in Figure 4c.

The averaged number of counter-ions and co-ions adsorbed in electrodes at applied and null potentials is presented in Table 3 to understand the trend of C_g with electrode types. Contrary to the trend observed in the variation of C_g with the electrode synthesis temperature, the number of adsorbed ions steadily increased as the synthesis temperature increased for both counter-ions and co-ions. An increasing number of adsorbed

ions indicated that the accessible volume increased with electrode synthesis temperature. Notably, in this context, the porosity of the electrodes did not change much with the synthesis temperatures, as shown in Table S1. The reduced accessible volume was due to small pores incapable of accommodating ions or narrow channels interconnecting the pores. The value of $C_g = 100.3 \text{ F/g}$ maximized for the CDC-1500 electrode despite the accessible volume increasing from the CDC-1000 to the CDC-2000 electrode. On the other hand, $C_g = 98.6 \text{ F/g}$ for CDC-1000 was only slightly below the maximum value, even though the number of adsorbed counter-ions was $\sim 17\%$ smaller than that for CDC-1500. The local structure of the electrodes had an impact on isolating and confining the ions, which, in turn, affected the efficacy of inducing charge in those ions. The degree of confinement and isolation was observed to have the opposite trend of accessible volume with respect to the electrode types, as discussed in a subsequent section. As a result of these competing trends, the capacitance was maximized for CDC-1500 with an intermediate pore size.

One way of characterizing the charging mechanism is through the X parameter, which typically varies from -1 to 1 .⁵² The boundary values of -1 and 1 correspond to electrodes charging by purely co-ion desorption and counter-ion adsorption, respectively. In the ion-exchange mechanism, where an equal number of counter-ions and co-ions is adsorbed and desorbed, the value for X is zero. This parameter has a continuous value because the charging mechanism may involve multiple mechanisms. For example, the counter-ion adsorption-dominated mechanism would involve counter-ions entering previously empty pores or adsorption of more counter-ions in places of desorbed co-ions due to the size discrepancy of ions. The charging mechanism parameter is defined by the following expression⁵³

$$X(V, 0) = \frac{[N_{\text{counter}}(V) + N_{\text{co}}(V)] - [N_{\text{counter}}(0) + N_{\text{co}}(0)]}{[N_{\text{counter}}(V) - N_{\text{co}}(V)] - [N_{\text{counter}}(0) - N_{\text{co}}(0)]} \quad (5)$$

where N_{counter} and N_{co} are the number of counter- and co-ions in pores, respectively.

The charging parameter X was calculated separately for the positive and negative electrodes due to the size difference in BMIM^+ and PF_6^- ions. The values of this parameter for all electrode types were between 0 and 1 , indicating that the charging mechanism involved counter-ion adsorption and ion exchange processes. In all electrode types, the X value for the positive electrode was higher than that for the negative electrode, as shown in Table 3. This discrepancy meant that the counter-ion adsorption was more dominant in the positive electrode than in the negative electrode. For CDC-1000 and

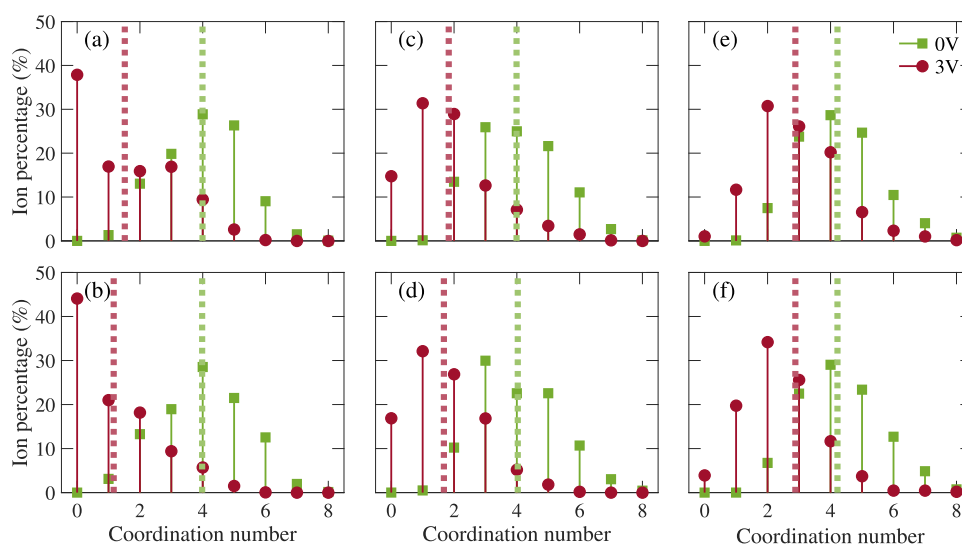


Figure 5. Coordination number distribution of counter-ions surrounded by co-ions within (a,c,e) positive and (b,d,f) negative electrodes for supercapacitors with (a,b) CDC-1000, (c,d) CDC-1500, and (e,f) CDC-2000 electrodes. The vertical lines represent the average coordination number of the profiles of the same color.

CDC-1500, the ion adsorption mechanism was the most dominant as ions entered empty pores. The ion exchange mechanism was diminished and enhanced in the positive and negative electrodes, respectively, due to the size difference between BMIM^+ and PF_6^- . In a negative electrode, a lesser number of counter-ions can occupy the volume left by the desorbed co-ions, as BMIM^+ was much larger than PF_6^- , thus reducing the ion adsorption mechanism and driving the X parameter down. The CDC-2000 electrode differed from the other electrode types, considering that most pore volumes were filled with ions at null potential. The ion-exchange mechanism was expected to dominate in this electrode, and it did for the negative electrode, as evidenced by the $X = 0.37$ value, significantly smaller than 0.5. However, in the positive electrode, more PF_6^- could enter the electrode left by the larger BMIM^+ ion, making the ion adsorption the dominant mechanism. As a result, the $X = 0.69$ parameter was similar to that of the CDC-1000 and CDC-1500 electrode types.

3.2. Local Environment Experienced by In-Pore Ions.

Figure 5 shows the coordination numbers at the null potential and an applied 3 V potential for different electrode types. The distributions at null potential had very similar patterns for all electrode types, with an average coordination number of ~ 4 in both positive and negative electrodes, as denoted by the dotted green lines in Figure 5. Notably, the peak coordination number in the bulk electrolyte was 6. This observation of partial desolvation conforms to a previous study that demonstrated the desolvation of counter-ions from solvent-free liquids as they were adsorbed in a porous carbon structure.⁵⁴

A leftward shift in the coordination number profile was observed in each plot at the applied voltage compared with the null potential. However, the extent of these leftward shifts differed for various electrode types, as is evident from the dotted red vertical lines that represent the average coordination number. The average coordination numbers for the negative electrodes were 1.14, 1.68, and 2.88 for CDC-1000, CDC-1500, and CDC-2000, respectively. This trend suggested that electrodes with smaller pores could desolvate the counter-ions more. A similar trend was observed for positive electrodes with slightly larger average coordination numbers. The variation in

the peak coordination number was identical for both the positive and negative electrodes. The values of the peak coordination number were 0, 1, and 2 for CDC-1000, CDC-1500, and CDC-2000, respectively. These observations indicated that when counter-ions were adsorbed in smaller pores, fewer co-ions could be present in the vicinity due to the confined space.

As the PSDs shifted to the right from CDC-1000 to CDC-2000, the volume of small pores decreased, while larger pores occupied more space. This reduction in the number of small pores resulted in the ions being less desolvated. To the authors' best knowledge, such a clear influence of electrode pore size on the degree of desolvation has not been reported in any previous studies. Merlet et al.³⁰ reported the coordination number profile for $\text{BMIM}^+\text{PF}_6^-$ solvated in acetonitrile. In that study, ions were categorized into four groups based on their DoC and coordination number profile calculated in relation to the solvent molecule. This profile was compared before and after voltage application for each group. However, a quantitative comparison with previously reported data is not feasible due to the presence of a solvent molecule. Nevertheless, a similar leftward shift in the coordination profile of counter-ions due to voltage application was also observed in the previous study.

A lower coordination number was conducive to inducing more charge on electrode atoms due to the fewer co-ions, which would otherwise diminish counter-ions' influence by the screening effect. CDC-1000 electrodes exhibited the lowest average coordination number of zero, with nearly 40% of the total adsorbed counter-ions desolvated. This characteristic allowed for charge accumulation in the CDC-1000 electrode, which was closer to that in the CDC-1500 electrode, even though the number of adsorbed counter-ions in CDC-1000 was about 19% lower compared to that in CDC-1500. The average coordination numbers for the positive and negative electrodes differed slightly between those of CDC-1000 and CDC-1500. Notably, the negative electrodes had a lower average coordination number than did the positive ones. The difference was due to the size variation of BMIM^+ and PF_6^- . The observation of the center of mass of the electrolytes revealed that the BMIM^+ ions were adsorbed more favorably closer to the pore walls. In contrast, PF_6^- ions were observed to occupy the

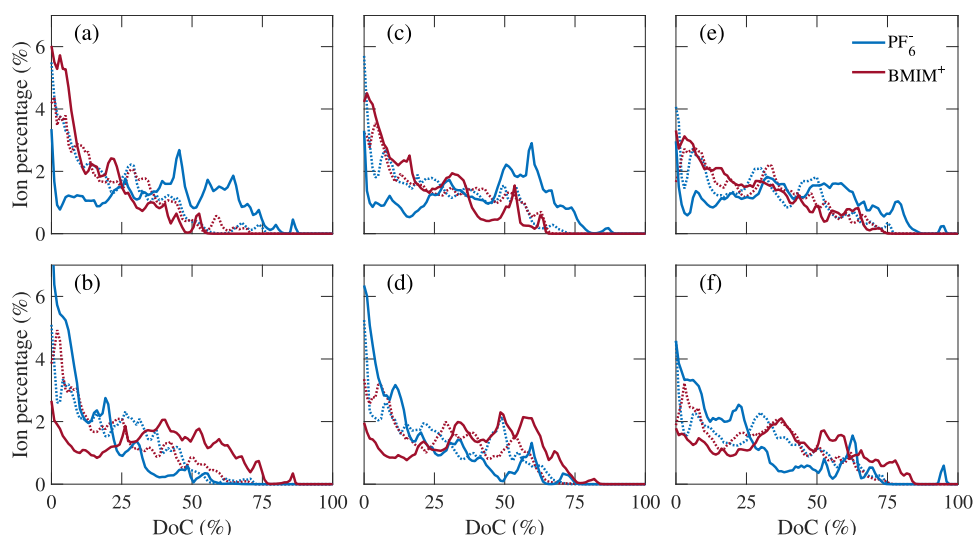


Figure 6. Comparison between the degree of confinement (DoC) of co-ions inside and in the vicinity of electrodes before and after voltage application to supercapacitors with (a,b) CDC-1000, (c,d) CDC-1500, and (e,f) CDC-2000 electrodes, where (a,c,e) are for positive electrodes and (b,d,f) are for negative electrodes. The broken and continuous lines represent DoC profiles before and after voltage application, respectively. The red and blue lines denote BMIM^+ and PF_6^- ions, respectively.

space at the center of the pores. Additionally, the C_1 and C_2 sites of BMIM^+ ion have a smaller σ value than that of the PF_6^- ion. These small σ values might allow the cations to be adsorbed in smaller confines, resulting in a slight increase of ion percentage in lower coordination numbers for negative electrodes.

At null potential, the DoC profiles of counter-ions and co-ions exhibited similar shapes in both positive and negative electrodes across all electrode types, extending up to 75% DoC, as illustrated in Figure 6. Once voltage was applied, the relative shift in the DoC profile remained consistent, regardless of electrode types and polarity. In each case, the percentage of atoms within the DoC range of <15–20% significantly increased for co-ions. However, this increase in the percentage did not correspond to a rise in the number of ions. Following the application of voltage, the total number of co-ions within the electrodes decreased for all electrode types, as shown in Figure 4. The high value observed in the low DoC region indicated that a substantial fraction of the co-ions was concentrated near the interface between electrode and electrolyte and in pores with low DoC. An additional common feature across all electrodes under applied voltage was the expulsion of co-ions from highly confined pores. For instance, as depicted in Figure 6a, the percentage of co-ions dropped to zero beyond the 56% DoC mark.

Previous studies have reported the DoC profiles for $\text{BMIM}^+\text{PF}_6^-$ solvated in acetonitrile³⁰ and pure 1-ethyl-3-methylimidazolium (EMIM) tetrafluoroborate (BF_4^-).³¹ Despite being based on different electrolytes, the DoC profiles in our work and those from previous studies share several similarities. One notable commonality is the high co-ion percentages observed within the DoC range of <15%. Furthermore, the DoC profile of $\text{EMIM}^+\text{BF}_4^-$ adsorbed in the CDC structure with an average pore size of 9.1 Å displayed distinct peaks in the range of 40% < DoC < 50%, for both positive and negative electrodes.³¹ In contrast, our CDC-1500 model, with an average pore size of 9.45 Å³⁷ exhibited counter-ion peaks in the range 50% < DoC < 60%.

The notable characteristic of the DoC profiles for the CDC-1000 electrode is that there is a zero percentage of co-ions

beyond 56% DoC at the applied potential, as illustrated in Figure 6a,b. In the CDC-1000 electrode, the pores with a DoC $\geq$ 56% were free of co-ions, with counter-ions occupying those spaces solely. As previously mentioned, the charging mechanism in the CDC-1000 electrode is primarily influenced by the adsorption of counter-ions, as indicated by the X values presented in Table 2. A portion of the newly adsorbed ions was located in previously unoccupied pores with high DoC. At null potential, ions accumulated in the pores near the electrode-bulk electrolyte interface. A portion of these co-ions, adsorbed in the high DoC pores, either migrated to lower DoC pores or were expelled from the electrode.

In high DoC regions, a similar difference in the profiles of counter-ions and co-ions was observed in the CDC-1500 and CDC-2000 electrodes, although to a lesser extent. Specifically, the percentages of co-ions dropped to zero for DoC values >65% for the CDC-1500 electrode and >75% for the CDC-2000 electrode, as illustrated in Figure 6c,e. The PF_6^- ion, being smaller than the BMIM^+ ion, was adsorbed more favorably to pores with higher DoC values. Consequently, a slight rightward shift was observed in the DoC profile of the positive electrode compared with that of the negative electrode. The change in the DoC profile across different electrode types can be quantified by the average DoC shift after the voltage is applied. For the negative CDC-1000 electrode, the average DoC values for the counter-ion were 20% at null potential and 35.4% when potential was applied, resulting in a shift of 15.4%. The DoC shifts for the CDC-1500 and CDC-2000 electrodes were 12.6% and 9.5%, respectively. Notably, the average DoC value for counter-ions at the applied voltage was similar for all electrodes, and the observed change in DoC resulted from differences in the average DoC values at null potential. Evidently, the electrode with a left-shifted PSD prevented electrolytes from occupying highly confined pores at the null potential.

The correlation between the average coordination number and the DoC is illustrated in Figure 7. The average DoC shows a decreasing trend with coordination numbers for both positive and negative electrodes. As the pores become more confined, there is less space available to accommodate co-ions. Notably,

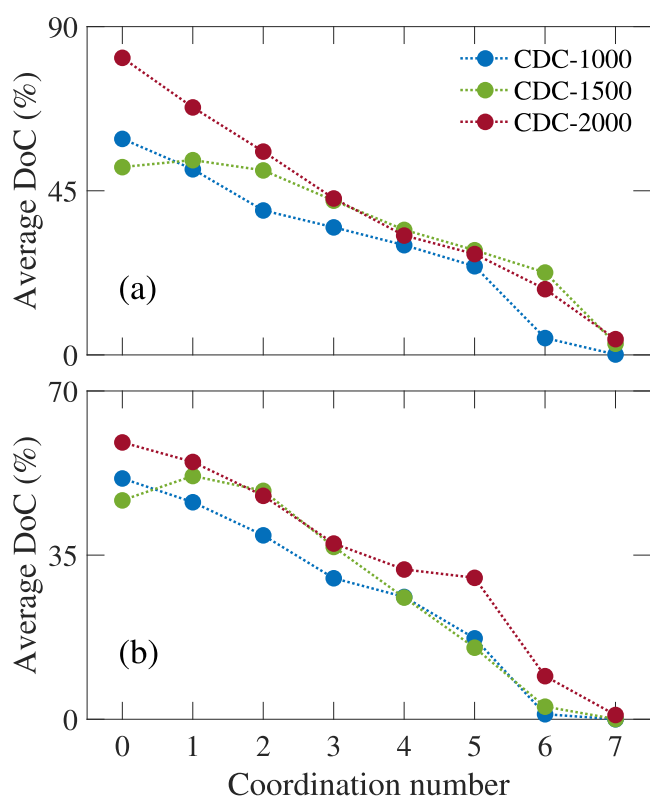


Figure 7. Variation of the average degree of confinement (DoC) with coordination number for counter-ions in (a) positive and (b) negative electrodes with CDC-1000, CDC-1500, and CDC-2000 and 3 V applied potential.

the average DoC value for the same coordination number is significantly higher for the CDC-2000 electrode than for the CDC-1000 electrode. This difference arises because a substantial portion of counter-ions occupies previously empty pores in the CDC-1000 electrode, allowing the ions to achieve lower coordination numbers even in a less confined space. In contrast, the CDC-2000 electrode was filled with co- and counter-ions at the null potential. As a result, the counter-ions must enter highly confined pores to reduce the number of cations surrounding them. Furthermore, the distribution of counter-ions in the CDC-1500 electrode differs from that of other types. In this case, the most confined counter-ions have a coordination number of 1, whereas the other electrodes exhibit a coordination number of zero. This behavior may be attributed to the size of the predominant pores in the CDC-1500 electrode, which can accommodate one counter-ion and one co-ion.

The parameter d_C is important for studying supercapacitors, because the capacitance is inversely proportional to the distance between opposite charges. The d_C profiles for all electrode types showed only slight deviations, indicating that the pore size did not significantly affect d_C . However, the profiles differed between positive and negative electrodes. Specifically, the d_C profiles for negative electrodes were more dispersed than those for positive electrodes, as depicted for the CDC-1000 electrode in Figure 8. The ϵ value of 1.12 kcal/mol for the PF_6^- ion was significantly higher than that for the BMIM^+ ion sites. Consequently, it was energetically favorable for the PF_6^- ion to be adsorbed in such a way that the minimum distance between the ion and the electrode atom reached the equilibrium distance. The BMIM^+ ion, which has three different charges and LJ parameters corresponding to three sites, experienced varying

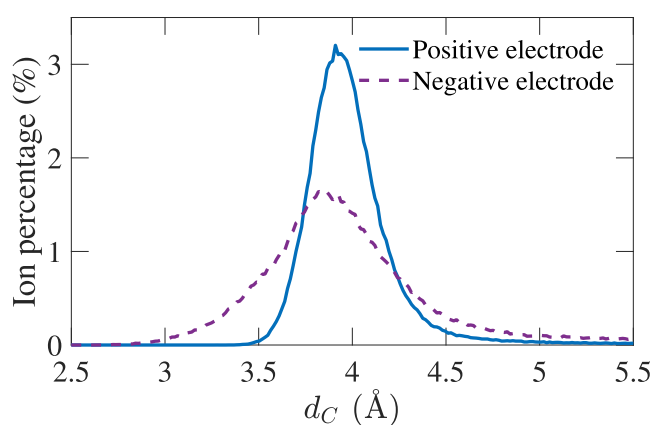


Figure 8. Distribution of the distance of closest approach (d_C) of counter-ions in the electrodes of the supercapacitor with the CDC-1000 electrode.

degrees of attraction to the pore walls, depending on its orientation, the structure of the pore walls, and the distribution of atomic charge on the wall atoms. Figure 10 shows the different orientations of the BMIM^+ ion relative to the nearest pore wall. These varied orientations contribute to a broad distribution of d_C . In contrast, PF_6^- is a single site ion, and its adsorption does not depend on orientation, resulting in a narrow profile of d_C .

The profile of d_C corresponding to the negative electrode exhibited a significantly high ion percentage in the small d_C region, even at <3.5 Å. In contrast, the profile for the positive electrode dropped to zero in this range, as illustrated in Figure 8. This notable presence of ions with small d_C values may seem counter-intuitive, given the larger size of the BMIM^+ ion, which has three adsorption sites, compared to the single site of the PF_6^- ion. Due to lower σ values, the C_1 and C_2 sites of BMIM^+ can approach the electrode atoms more closely than the C_3 site or the PF_6^- ion. Because the center of mass for BMIM^+ is nearest to the C_1 site, this results in a small d_C for the BMIM^+ ion, even <3.5 Å. The varying local structures of the pores allow for different extents of the closest approach for BMIM^+ ions at the negative electrode, as discussed in Section 3.4.

3.3. Atomic Charge Distribution of Electrodes. The cations occupied pores with a higher DoC as the electrode synthesis temperature increased. A higher percentage of co-ions reduces the surface area available for counter-ion adsorption, which in turn diminishes the useful charge induced on electrode atoms. The profiles of DoC and coordination number indicated that the effectiveness of counter-ions in inducing charge was reduced in electrodes with a right-shifted PSD. We binned the number of electrode atoms according to their atomic charge and normalized this profile against the total number of carbon atoms in the electrode. Figure 9 illustrates the distributions obtained by averaging 2000 frames collected from the last 10 ns of the simulation. The distributions varied between positive and negative electrodes due to the different shapes of the adsorbed electrolyte ions. Despite these differences in profiles, the mean atomic charge calculated from the distributions was equal for both positive and negative electrodes, thereby maintaining charge neutrality. The mean atomic charge followed the same trend as the steady-state charge, which was measured at 0.0178, 0.0186, and 0.0165 e per atom for CDC-1000, CDC-1500, and CDC-2000, respectively. Notably, the profiles corresponding to the CDC-2000 electrode exhibited a significant negative tail for positive electrodes and a positive tail for negative electrodes.

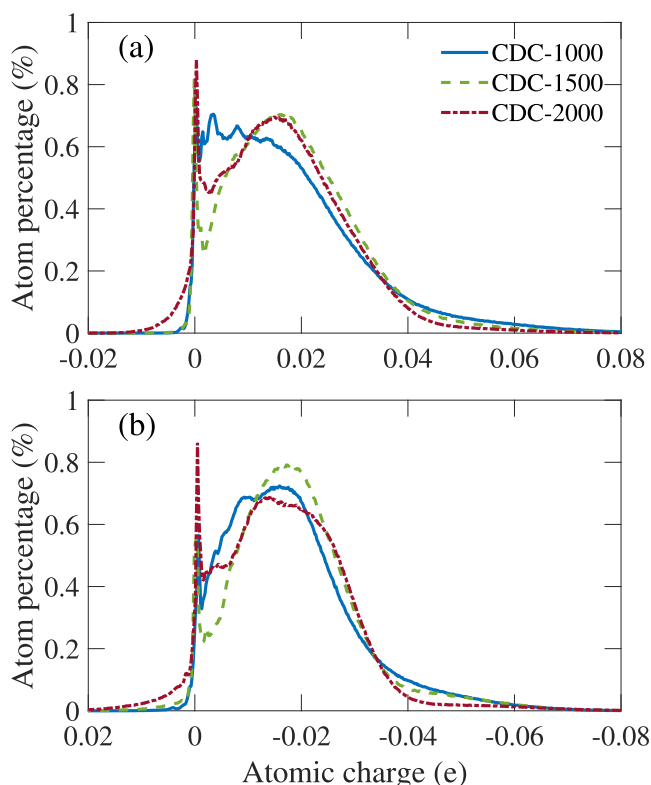


Figure 9. Distribution of atomic charge of electrode atoms in (a) positive and (b) negative electrodes with CDC-1000, CDC-1500, and CDC-2000 at the applied potential of 3 V.

The large number of co-ions was responsible for inducing charges that mirrored the polarity of the applied potential. A common feature in all distributions was the sharp peak at zero atomic charge, attributed to the carbon atoms located in the inaccessible pores.

For the CDC-1000 electrode, the atom percentage was considerably higher than those of other electrodes in the atomic charge range from zero to $\sim 0.011e$ ($-0.015e$) in the positive (negative) electrode. However, the percentage dropped below other profiles beyond $\sim 0.011e$ ($-0.011e$). The profiles corresponding to CDC-2000 had the opposite characteristics of CDC-1000, with a lower atom percentage from zero to $\sim 0.011e$ ($-0.011e$) and a higher percentage beyond $\sim 0.011e$ ($-0.02e$) in the positive (negative) electrode, compared to

CDC-1000. However, the profiles for CDC-2000 also had a significant atom percentage below (above) zero atomic charge for positive (negative) electrodes. The presence of oppositely charged atoms reduced the mean atomic charge value of the CDC-2000 electrode despite having a higher atom percentage in the high atomic charge range in comparison to that of CDC-1000. The profile of the CDC-1500 electrode had characteristics similar to those of both CDC-1000 and CDC-2000 electrodes. In the atomic charge region above (below) $0.011e$ ($-0.02e$), the CDC-1500 profile closely followed that of CDC-2000 for the positive (negative) electrode. On the other hand, the CDC-1500 profile followed the same pattern as CDC-1000 below (above) zero charge in the positive (negative) electrode. Thus, the CDC-1500 electrode had the highest mean atomic charge.

3.4. Case Studies on Ions Adsorbed in Different Local Environments. Figure 10a illustrates the scenario of an ion being adsorbed onto the electrode, where the distance d_c is 3.48 Å. The ion was adsorbed at the edge of a graphitic sheet, achieving a minimum distance of 2.96 Å between the C_2 site and the electrode atom. The close proximity of the adsorbed ions to the edge site resulted in a significant atomic charge of ($\sim 0.053e$) on the carbon atom, indicated by the deep blue color. Many edge sites exhibited such high charges; however, not all edge atoms acquired high atomic charges due to factors like the presence of co-ions, which also affected charge accumulation. The snapshot corresponding to $d_c = 3.94$ Å is shown in Figure 10b. In this case, the ion was adsorbed within a pore that had a high DoC, surrounded by a curved pore wall. The highest percentage of ions, as indicated in Figure 8 for the negative electrode at this distance, showed that this local structure was the most common adsorption site for the BMIM⁺ ions. Conversely, the less confined ions were loosely adsorbed to the pore walls, resulting in higher distances, as demonstrated in Figure 10c, where $d_c = 4.53$ Å. The local environment, in this case, resembled a slit pore structure formed by two nearly planar graphene sheets. The capacitance is inversely proportional to the distance between the adsorbed ion and the electrode pore wall. As a result, d_c is an essential parameter regarding electrolyte selection for optimizing device performance.

The counter-ions were expected to be adsorbed in larger pores to a greater degree as the average pore size of an electrode increased. The counter-ion occupying a pore was defined by the largest sphere that encompassed the ion and also did not overlap with any carbon atom. After defining the pores, the percentage of counter-ions adsorbed in pores with certain radii was calculated as detailed in the Supporting Information. For positive

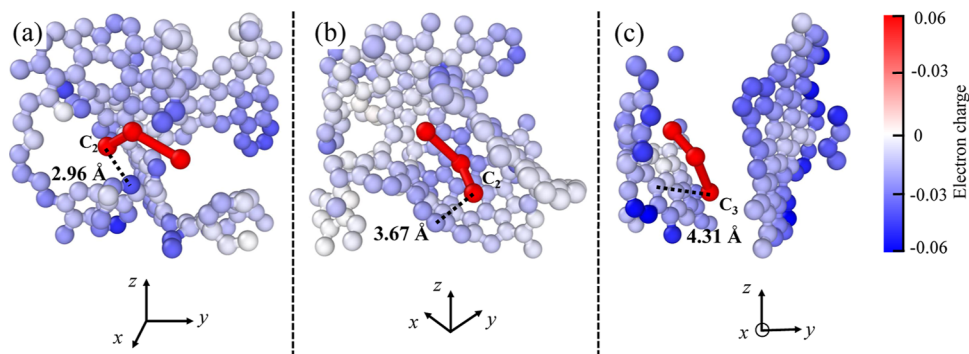


Figure 10. Snapshots of the cross-sectional views of the negative electrode showing electrode atoms and only the ions of interest of the CDC-1000 electrode, exhibiting BMIM⁺ ions adsorbed with varying degrees of proximity. The distances of the closest approach (d_c) for the adsorbed ions are (a) 3.48 Å, (b) 3.94 Å, and (c) 4.53 Å. The dotted black lines show the smallest distance between the electrode atom and cation sites.

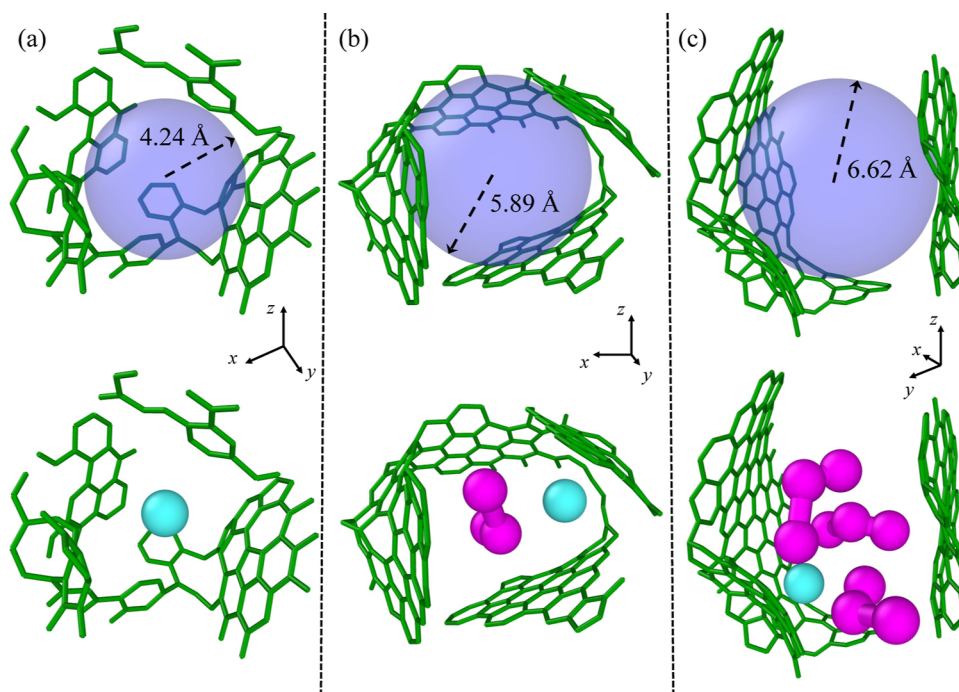


Figure 11. Schematic diagrams of cross sections of top electrodes for supercapacitors with three different electrode types: (a) CDC-1000, (b) CDC-1500, and (c) CDC-2000. Each pair of figures represents the same cross-section. The top figure displays a sphere fitted to the carbon structure for defining the pore size, while the bottom figure shows the counter-ion and co-ions contained within the specified pore. The pores, along with the BMIM⁺ and PF₆[−] ions, are represented by semitransparent blue, magenta, and cyan spheres, respectively. The radii of the pores are denoted in the figures.

electrodes, the counter-ions predominately adsorbed in pores with radii in the interval defined by 4–4.5 Å, 5.5–6 Å, and 6.5–7 Å for CDC-1000, CDC-1500, and CDC-2000, respectively. The electrode cross-section, including the carbon atoms, counter-ions, and co-ions, is presented in Figure 11 to study the impact of pore size on charge accumulation on the pore wall. In the cases shown in Figure 11, the pores represented by the blue spheres have radii of 4.24, 5.89, and 6.62 Å for CDC-1000, CDC-1500, and CDC-2000, respectively. The charge induced by the counter-ions is 2.58, 1.7, and 1.39 for CDC-1000, CDC-1500, and CDC-2000, respectively. The high value for the case of CDC-1000 can be attributed to two observations. First, the counter-ion had no co-ion inside the sphere of influence, as shown in Figure 11a. Moreover, the counter-ion was tightly surrounded by carbon pore walls, as evidenced by the high DoC value of 67.3%. On the contrary, the coordination numbers for CDC-1500 and CDC-2000 were 1 and 3, as can be seen from Figure 11b,c. In these cases, the counter-ions were adsorbed at one side of the pores as the pores were much bigger than that of CDC-1000. As a result, the ions could only induce charge on parts of the carbon atoms constituting the pore walls. Screening effect from co-ions and influencing small pore segments resulted in the total induced charge being less in these scenarios. The lesser charge for CDC-2000 can be attributed to the presence of three counter-ions compared to only one in CDC-1500.

4. CONCLUSION

Our work utilized structurally realistic atomistic models of the CDC porous structure, which more accurately matched the experimental samples compared to previous models. We simulated the EDLC device with an RTIL electrolyte by using the CPM method. The transient charging profile displayed a biexponential behavior, where the pore distribution of the

structures affected the time constants. Larger pores allowed the ions to diffuse more rapidly throughout the structure, resulting in smaller time constants. The local environment experienced by adsorbed electrolyte ions significantly influenced their ability to induce charges on the electrode atoms. In the CDC structures with smaller average pore sizes, the isolation and adsorption of counter-ions were stronger, as evidenced by coordination numbers and the distribution of DoC. As the temperature of electrode synthesis increased, the number of pores inaccessible to ions decreased, allowing more ions to enter the electrodes. Therefore, the CDC-1500 electrode showed a greater specific capacitance than the CDC-1000 electrode as the steady-state charge of the CDC-1500 electrode was greater than that of the CDC-1000 electrode. However, the reduced capacity of the isolating counter-ions and co-ions offsets the benefit of the even larger accessible volume, resulting in a decrease in the steady-state charge of the CDC-2000 electrode compared to that of the CDC-1500 electrode.

These insights will be crucial for modeling supercapacitor behavior and optimizing the performance. Although incorporating realistic electrode materials is a significant advancement toward accurate modeling of supercapacitors, further development of the simulation methodology is needed. The current model does not account for the impact of the density of states of the electrode atoms and the polarizability of electrolyte ions. The accuracy of our simulation can be enhanced by incorporating a model that considers the effects of quantum capacitance, specifically the influence of the density of states of the electrode atoms.

■ ASSOCIATED CONTENT

Data Availability Statement

All data in the paper are present in the main text, which will also be available from the corresponding author upon reasonable request.

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jpcc.5c03167>.

Structural parameters of the atomistic model of the electrodes and a detailed description of the methodology for calculating the solid angle employed in this work (PDF)

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Notes

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■ ABBREVIATIONS

BMIM 1-butyl-3-methylimidazolium
CDC carbide-derived carbon
CPM constant potential method
DoC degree of confinement
EDLC electric double layer capacitor
LJ Lennard-Jones
MD molecular dynamics
PF₆ hexafluorophosphate
PSD pore size distribution
RDF radial distribution function
RTIL room temperature ionic liquid

■ REFERENCES

- (1) Zhao, J.; Burke, A. F. Electrochemical capacitors: Materials, technologies and performance. *Energy Storage Mater.* **2021**, *36*, 31–55.
- (2) Zhang, L.; et al. A review of supercapacitor modeling, estimation, and applications: A control/management perspective. *Renewable Sustainable Energy Rev.* **2018**, *81*, 1868–1878.
- (3) Mitali, J.; Dhinakaran, S.; Mohamad, A. Energy storage systems: A review. *Energy Storage Sav.* **2022**, *1*, 166–216.
- (4) Pegueroles-Queralt, J.; Bianchi, F. D.; Gomis-Bellmunt, O. A power smoothing system based on supercapacitors for renewable distributed generation. *IEEE Trans. Ind. Electron.* **2015**, *62*, 343–350.
- (5) Partridge, J.; Abouelamaimen, D. I. The role of supercapacitors in regenerative braking systems. *Energies* **2019**, *12*, 2683.
- (6) Muzaffar, A.; Ahamed, M. B.; Deshmukh, K.; Thirumalai, J. A review on recent advances in hybrid supercapacitors: Design,

fabrication and applications. *Renewable Sustainable Energy Rev.* **2019**, *101*, 123–145.

(7) Hashemi, M.; et al. The use of an electrocatalytic redox electrolyte for pushing the energy density boundary of a flexible polyaniline electrode to a new limit. *Nano Energy* **2018**, *44*, 489–498.

(8) Xiong, G.; Kundu, A.; Fisher, T. S. *Thermal effects in supercapacitors*; Springer, 2015; Vol. 10.

(9) Ali, F.; Liu, X.; Zhou, D.; Yang, X.; Xu, J.; Schenk, T.; Müller, J.; Schroeder, U.; Cao, F.; Dong, X. Silicon-doped hafnium oxide anti-ferroelectric thin films for energy storage. *J. Appl. Phys.* **2017**, *122*, 144105.

(10) Zhong, C.; et al. A review of electrolyte materials and compositions for electrochemical supercapacitors. *Chem. Soc. Rev.* **2015**, *44*, 7484–7539.

(11) Wang, Y.; Liu, L.; Ma, T.; Zhang, Y.; Huang, H. 2D graphitic carbon nitride for energy conversion and storage. *Adv. Funct. Mater.* **2021**, *31*, 2102540.

(12) Asen, P.; Shahrokhian, S. A high performance supercapacitor based on graphene/polypyrrole/Cu₂O–Cu(OH)₂ ternary nanocomposite coated on nickel foam. *J. Phys. Chem. C* **2017**, *121*, 6508–6519.

(13) Li, T.; et al. High-energy asymmetric electrochemical capacitors based on oxides functionalized hollow carbon fibers electrodes. *Nano Energy* **2016**, *30*, 9–17.

(14) Zhang, L. L.; Zhao, X. Carbon-based materials as supercapacitor electrodes. *Chem. Soc. Rev.* **2009**, *38*, 2520–2531.

(15) Chmiola, J.; et al. Anomalous increase in carbon capacitance at pore sizes less than 1 nm. *Science* **2006**, *313*, 1760–1763.

(16) Largeot, C.; et al. Relation between the ion size and pore size for an electric double-layer capacitor. *J. Am. Chem. Soc.* **2008**, *130*, 2730–2731.

(17) Prehal, C.; Koczwar, C.; Jäkel, N.; Schreiber, A.; Burian, M.; Amenitsch, H.; Hartmann, M. A.; Presser, V.; Paris, O. Quantification of ion confinement and desolvation in nanoporous carbon supercapacitors with modelling and in situ X-ray scattering. *Nat. Energy* **2017**, *2*, 16215.

(18) Zhao, C.; et al. CP/C = O bonds assisted desolvation effect in ultra-micropores carbon for boosting Zn-ion storage capability. *Energy Storage Mater.* **2023**, *58*, 332–343.

(19) Forse, A. C.; et al. NMR study of ion dynamics and charge storage in ionic liquid supercapacitors. *J. Am. Chem. Soc.* **2015**, *137*, 7231–7242.

(20) Forse, A. C.; Griffin, J.; Merlet, C.; Carretero-Gonzalez, J.; Raji, A. R.; Trease, N.; Grey, C. Direct observation of ion dynamics in supercapacitor electrodes using in situ diffusion NMR spectroscopy. *Nat. Energy* **2017**, *2*, 16216.

(21) Merlet, C.; et al. Simulating supercapacitors: can we model electrodes as constant charge surfaces? *J. Phys. Chem. Lett.* **2013**, *4*, 264–268.

(22) Harris, P. J. Structure of non-graphitising carbons. *Int. Mater. Rev.* **1997**, *42*, 206–218.

(23) Liu, Y. M.; Merlet, C.; Smit, B. Carbons with regular pore geometry yield fundamental insights into supercapacitor charge storage. *ACS Cent. Sci.* **2019**, *5*, 1813–1823.

(24) Pean, C.; et al. Confinement, desolvation, and electrosorption effects on the diffusion of ions in nanoporous carbon electrodes. *J. Am. Chem. Soc.* **2015**, *137*, 12627–12632.

(25) Harris, P. J.; Tsang, S. C. High-resolution electron microscopy studies of non-graphitizing carbons. *Philos. Mag. A* **1997**, *76*, 667–677.

(26) Acharya, M.; et al. Simulation of nanoporous carbons: a chemically constrained structure. *Philos. Mag. B* **1999**, *79*, 1499–1518.

(27) Feng, G.; et al. The importance of ion size and electrode curvature on electrical double layers in ionic liquids. *Phys. Chem. Chem. Phys.* **2011**, *13*, 1152–1161.

(28) Palmer, J.; et al. Modeling the structural evolution of carbide-derived carbons using quenched molecular dynamics. *Carbon* **2010**, *48*, 1116–1123.

(29) Merlet, C.; et al. On the molecular origin of supercapacitance in nanoporous carbon electrodes. *Nat. Mater.* **2012**, *11*, 306–310.

- (30) Merlet, C.; et al. Highly confined ions store charge more efficiently in supercapacitors. *Nat. Commun.* **2013**, *4*, 2701.
- (31) Pak, A. J.; Hwang, G. S. Molecular insights into the complex relationship between capacitance and pore morphology in nanoporous carbon-based supercapacitors. *ACS Appl. Mater. Interfaces* **2016**, *8*, 34659–34667.
- (32) De Tomas, C.; et al. Structural prediction of graphitization and porosity in carbide-derived carbons. *Carbon* **2017**, *119*, 1–9.
- (33) Van Duin, A. C.; Dasgupta, S.; Lorant, F.; Goddard, W. A. ReaxFF: a reactive force field for hydrocarbons. *J. Phys. Chem. A* **2001**, *105*, 9396–9409.
- (34) Lin, X.; Tee, S. R.; Searles, D. J.; Cummings, P. T. Molecular insights on optimizing nanoporous carbon-based supercapacitors with various electrolytes. *Electrochim. Acta* **2024**, *474*, 143500.
- (35) Thompson, M. W.; Dyatkin, B.; Wang, H. W.; Turner, C.; Sang, X.; Unocic, R.; Iacovella, C.; Gogotsi, Y.; Van Duin, A.; Cummings, P. An atomistic carbide-derived carbon model generated using Reaxff-based quenched molecular dynamics. *C* **2017**, *3*, 32.
- (36) Zhou, X.; Ward, D. K.; Foster, M. E. An analytical bond-order potential for carbon. *J. Comput. Chem.* **2015**, *36*, 1719–1735.
- (37) Sarkar, K.; Talukder, M. A. Structurally realistic carbide-derived carbon model in annealing molecular dynamics methodology with analytic bond-order potential. *Mater. Adv.* **2024**, *5*, 5738–5748.
- (38) Marks, N.; Cover, M.; Kocer, C. Simulating temperature effects in the growth of tetrahedral amorphous carbon: the importance of infrequent events. *Appl. Phys. Lett.* **2006**, *89*, 131924.
- (39) Steele, W. A. The interaction of rare gas atoms with graphitized carbon black. *J. Phys. Chem.* **1978**, *82*, 817–821.
- (40) Sun, H.; Ren, P.; Fried, J. The COMPASS force field: parameterization and validation for phosphazenes. *Comput. Theor. Polym. Sci.* **1998**, *8*, 229–246.
- (41) Roy, D.; Maroncelli, M. An improved four-site ionic liquid model. *J. Phys. Chem. B* **2010**, *114*, 12629–12631.
- (42) Plimpton, S. Fast parallel algorithms for short-range molecular dynamics. *J. Comput. Phys.* **1995**, *117*, 1–19.
- (43) Pollock, E.; Glosli, J. Comments on P³M, FMM, and the Ewald method for large periodic Coulombic systems. *Comput. Phys. Commun.* **1996**, *95*, 93–110.
- (44) Yeh, I.-C.; Berkowitz, M. L. Ewald summation for systems with slab geometry. *J. Chem. Phys.* **1999**, *111*, 3155–3162.
- (45) Andersen, H. C. Rattle: A “velocity” version of the shake algorithm for molecular dynamics calculations. *J. Comput. Phys.* **1983**, *52*, 24–34.
- (46) Ryckaert, J.-P.; Ciccotti, G.; Berendsen, H. J. Numerical integration of the cartesian equations of motion of a system with constraints: molecular dynamics of n-alkanes. *J. Comput. Phys.* **1977**, *23*, 327–341.
- (47) Siepmann, J. I.; Sprik, M. Influence of surface topology and electrostatic potential on water/electrode systems. *J. Chem. Phys.* **1995**, *102*, 511–524.
- (48) Ahrens-Iwers, L. J.; Janssen, M.; Tee, S. R.; Meißner, R. H. ELECTRODE: An electrochemistry package for atomistic simulations. *J. Chem. Phys.* **2022**, *157*, 084801.
- (49) Reed, S. K.; Lanning, O. J.; Madden, P. A. Electrochemical interface between an ionic liquid and a model metallic electrode. *J. Chem. Phys.* **2007**, *126*, 084704.
- (50) Noh, C.; Jung, Y. Understanding the charging dynamics of an ionic liquid electric double layer capacitor via molecular dynamics simulations. *Phys. Chem. Chem. Phys.* **2019**, *21*, 6790–6800.
- (51) Shi, Y. A mimetic porous carbon model by quench molecular dynamics simulation. *J. Chem. Phys.* **2008**, *128*, 234707.
- (52) Griffin, J. M.; et al. Ion counting in supercapacitor electrodes using NMR spectroscopy. *Faraday Discuss.* **2014**, *176*, 49–68.
- (53) Forse, A. C.; Merlet, C.; Griffin, J. M.; Grey, C. P. New perspectives on the charging mechanisms of supercapacitors. *J. Am. Chem. Soc.* **2016**, *138*, 5731–5744.
- (54) Ania, C. O.; et al. Polarization-induced distortion of ions in the pores of carbon electrodes for electrochemical capacitors. *Carbon* **2009**, *47*, 3158–3166.